Chapter One

Introduction

Compiled by Yilu

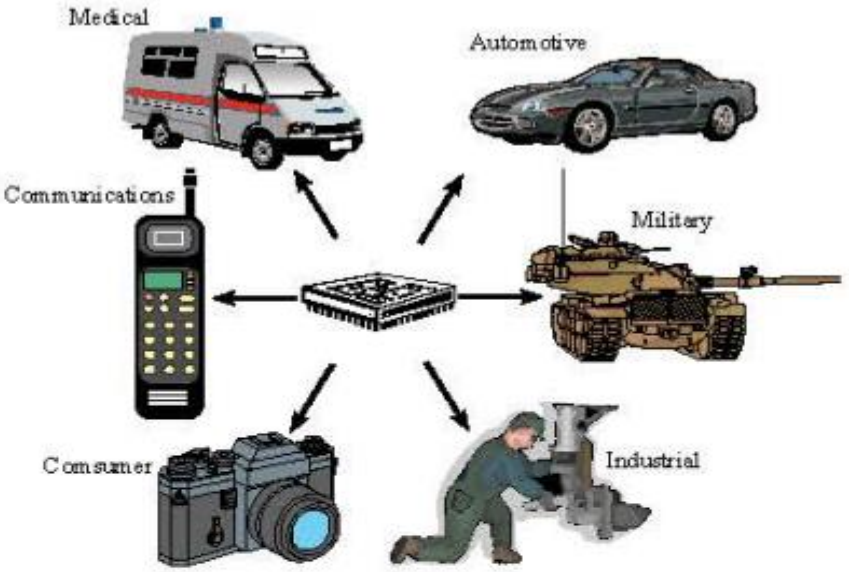
Module Outline

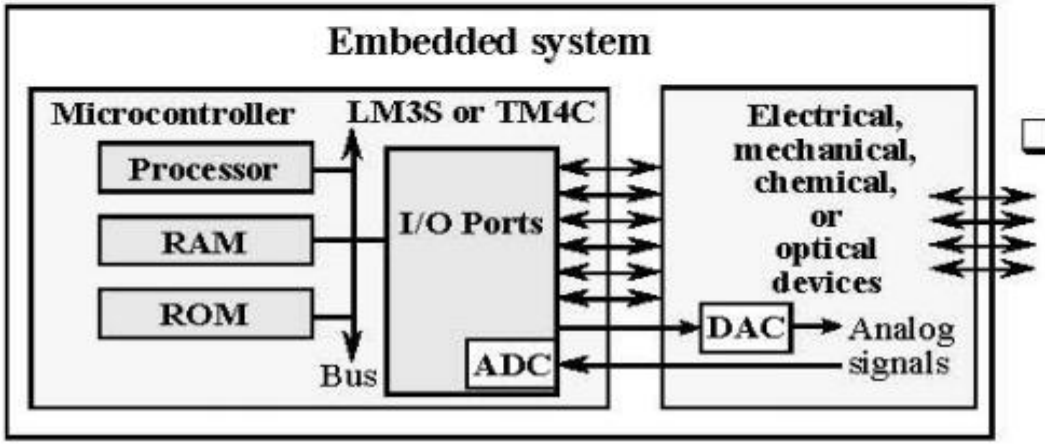
- ✓ *Definitions, characteristics and examples of real-time and embedded systems*
- ✓ *Model of real time system*
- ✓ *Types of real time tasks*
- ✓ *Modeling timing constraints*
- ✓ *Design process*

Definitions, characteristics and examples of real-time and embedded systems

- **What is the general purpose computer system?**
  - A computer system that can do **many functionalities** depend on available hardware and installed software. E.g., PCs, Server
- **What makes a system Embedded?**
  - ✓ **An embedded system** is a computer system designed to perform one or few dedicated function often with real time computing constraints.
  - ✓ It is a microcomputer system embedded in a larger system and designed for one or two dedicated services.
  - ✓ It is embedded as part of a complete device that often has hardware and mechanical parts.
  - ✓ Real-time systems, however, are required to compute and deliver correct results within a **specified period of time**.

Cntd…





- Embedded Systems are everywhere
  - Ubiquitous, invisible
  - Hidden (computer inside)
  - Dedicated purpose
- Microprocessor
  - Intel: 4004, ..8080,.. x86
  - Freescale: 6800, .. 9S12,.. PowerPC
  - ARM, DEC, SPARC, MIPS, PowerPC, Natl. Semi.,…
- Microcontroller
  - Processor+Memory+ I/O Ports (Interfaces)

Source:
Erez, Gerstlauer, Janapa Reddi, Telang, Tiwari, Valvano, Yerraballi

Cntd…

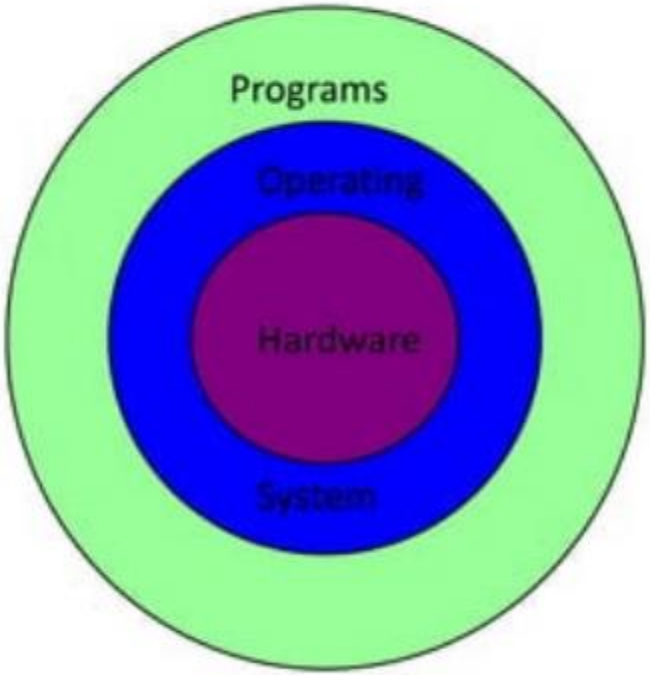
- **HW +SW**

Embedded systems (ES) = information processing systems embedded into a larger product

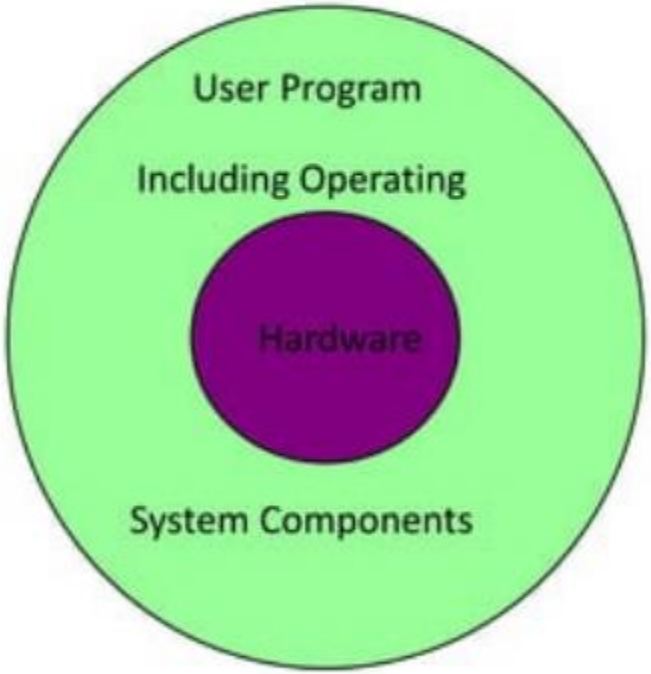
- The embedded system is designed to **run on its own with out human intervention** and may be required to respond to events in real time.
- An application specific electronic sub system which is completely encapsulated but the main system it belongs to:
  - Designed for some specific tasks
  - Subjected to real time performance constraints that must be met
  - Feature tightly integrated combinations of hardware and software

Cntd…

Embedded system Configuration



Typical Computer system Configuration



Typical Embedded system Configuration

Comparison

General purpose Computing

- Broad class of application
- Programmable by end user
- Faster is better
- **Criteria**
  - More complex circuit
  - High Cost
  - Require more memory
  - Use high speed processor
  - More complex circuit

Embedded System

- Few application that are known at design time
- Not programmable by end user
- Fixed run time requirements (additional computing power not useful)
- **Criteria**
  - Low Cost
  - Low power consumption
  - Predictable
  - Less memory
  - Use slow processor
  - Less memory is required

Real-Time Embedded System Characteristics

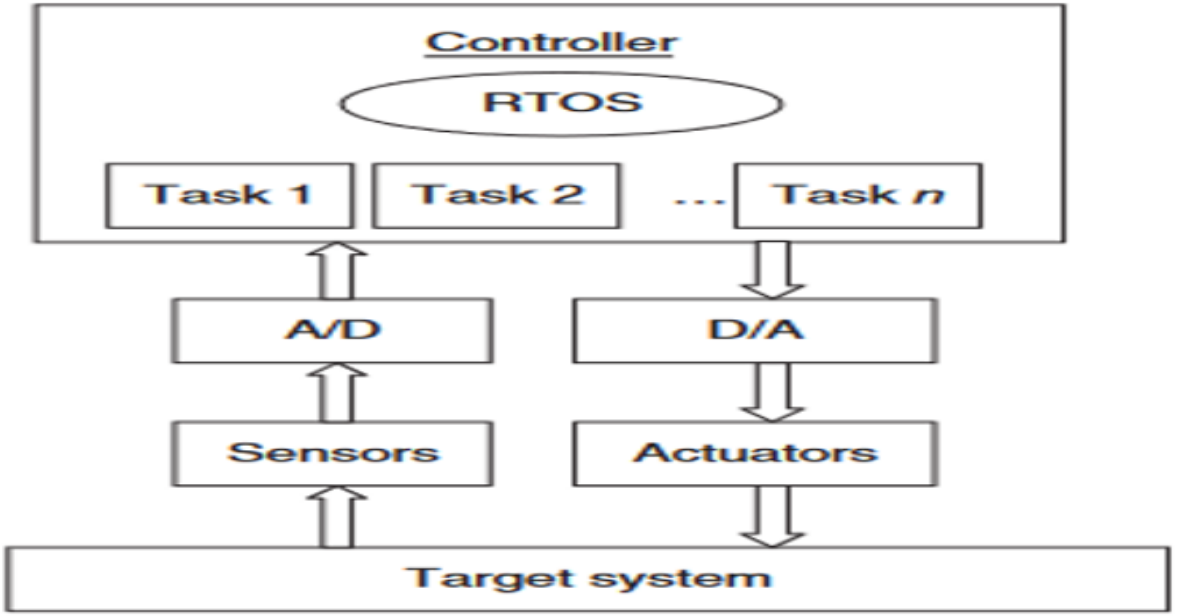
- The following are basic characteristics of real time and embedded system
  - System Structure
  - Real time resource
  - Highly Constrained environment
  - Concurrency
  - Predictability
  - Safety and Reliability

1. System Structure

- A real time embedded system interacts with **its environment** continuously and timely.
- **Sensors** are used to control or monitor the environment
- Most of the data from the physical environment are in the form of the analog Signals.
  - Analog need to converted to digital to manipulate the data using the micro processor.
  - Need to be converted to digital signals:- an analog- to- digital converter (ADC) is needed in between **a sensor** and **microprocessor**.

Cntd…

- ***Controller*** *is the brain of the embedded system*
  - ***Controller****= one or microprocessor + memory + Some peripherals+ a real time software application*
  - ***The controller*** *acts through actuators*
    - *Need to convert digital signal to analog, hence DAC is needed to be preformed in order to apply the microprocessor output to the actuator.*



Fig, Structure of real time embedded system

Cntd…

2. **Real Time Response**

- Has to finish certain tasks within specific time boundaries
- Real time systems are expected to execute all tasks by their **deadline.**
  - Deadline of real time tasks are typically derived from the required responsiveness of the **sensors**, **actuators** and the dynamics of the target that the embedded system controls.

3. **Highly Constrained Environments**

- Constrained in terms of processor speed ,memory capacity and user interface
  - Ina the realm of IoT power constraint is included.
- Real time embedded systems are often run in highly resource constrained environments, which makes the system design and performance optimization quite **challenging.**

Cntd…

- Therefore, embedded system must be optimized in terms of
  - Size and weight –easy to mount
  - Reliability – deadlines and critical ness
  - Performance-service on time but not necessarily fast + efficient use of Limited memory
  - Cost- affordability
  - Power consumption- remote deployment

3. Concurrency:

- It refers to a property of systems in which several computations **(tasks)** are executing simultaneously and potentially interacting with each other.
- Almost all tasks have strict constraints on the response time – **deadline**
  - All deadlines should be met

Cntd…

4. Predictability

- A real- time system must be have in a way that can be predicated in terms of all timing requirements.
- Factors to prediction
  - Workload,
  - Processor speed,
  - Real time operating system support
  - Process and thread priorities
  - Scheduling algorithm
  - Communication infrastructure and so on.

Cntd…

5. Safety and Reliability

- Safety: critical real time embedded systems must highly reliable

  E.g., cardiac pacemakers, flight control systems, elevators.

- **Safety:**
  - "freedom from accidents or losses"
  - Usually concerned with safety in the absence of faults as well as the presence of single point of faults.
- **Reliability:**
  - Refers the ability of system or component to performs its required functions under stated conditions for a specific time
  - Measured in failures per million operating hours. E.g., the requirements for a typical automotive microcontroller is 0.12 failures per million operating hours.

Application Area

- The main system can range from **personal** to **industrials**
- household appliance, home automation, consumer electronics, ATMs, network routers, automobiles etc.

- Consumer electronics:
  - Washing machine, Exercise equipment, Remote controls, Clocks and watches, Games and toys, Audio/video electronics, Set-back thermostats, Camera, Camcorder, Television, VCR, cable box
- Communication systems:
  - Answering machines, Telephones, Fax machines, Radios, Cellular phones, pagers
- Automotive systems:
  - Automatic breaking, Noise cancellation, Locks, Electronic ignition, Power windows and seats, Cruise control, Collision avoidance, Climate control, Emission control, Instrumentation
- Military hardware:
  - Smart weapons, Missile guidance systems, Global positioning systems, Surveillance systems
- Business applications:
  - Cash registers, Vending machines, ATM machines, Traffic controllers, Industrial robots, Bar code readers and writers, Automatic sprinklers, Elevator controllers, RFID systems, Lighting and heating systems
- Medical devices:
  - Monitors, Drug delivery systems, Cancer treatments, Pacemakers, Prosthetic devices, Dialysis machines
- … and more

Real Time Systems

- **Real-time systems** defined as: "those systems in which the correctness of the system depends not only on the **logical result of the computation**, but also **on the time** at which the results are produced

- *Real-time*
  - *Does not denote speed/fast*
  - *But meet the timing constraints or deadlines*
- ***Goal:*** *make the system predictable and robust*
- ***Predictability:*** *have deterministic behavior*
  - *Usually use the worst-case analysis*
  - *Guaranteed response/reaction times*
- ***Robustness:*** *the job with granted performance will not be interfered with other job*

Characteristics of Real-Time Systems

- **Characteristics**
  - Must produce correct computational results, called *logical or functional correctness*
  - These computations must conclude within a predefined period, called *timing correctness*
- The overall correctness of a real-time system depends on both the *functional correctness* and the *timing correctness*

Type of Real-Time System

- ***Hard Real-Time:***

➢ **A hard real-time system** is one in which failure to meet even a single deadline may lead to **complete or catastrophic system failure**.

➢ Degree of tolerance of missed deadline

Extremely small or zero

➢ Usefulness of computed results after missed deadlines

Likely useless

➢ Severity of the penalty incurred for failing to meet deadlines

catastrophe

E.g. Nuclear reactors, Flight controller , Weapon defense system, Missile guidance system

Cntd….

- *Soft Real Time*

- **A soft real-time system** is one in which performance is degraded but not destroyed by failure to meet response-time constraints
- Degree of tolerance of missed deadline

  Non-zero
- Usefulness of computed results after missed deadlines

  Have a rate of depreciation
- Severity of the penalty incurred for failing to meet deadlines

  Non-catastrophic
- **E.g.:** Multimedia Streaming, Electronic games, Quality-of-Service (QoS) guarantees

Cntd…

Firm Real Time

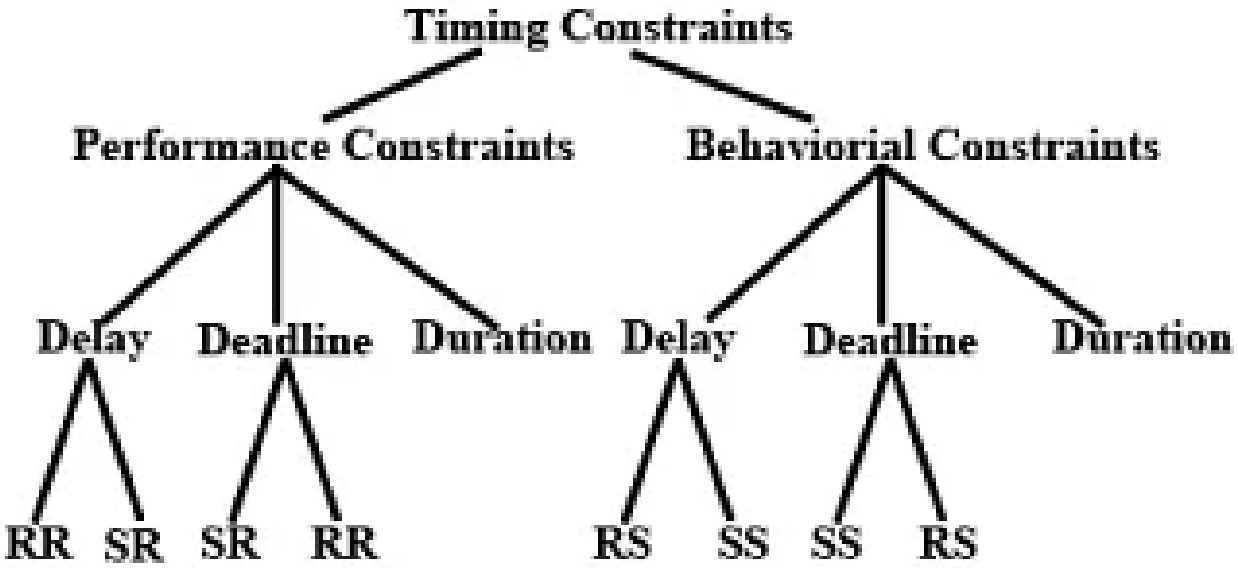
- A firm real-time system is one in which a few missed deadlines will not lead to total failure, but missing more than a few may lead to complete or catastrophic system failure.
- Missing a few navigation deadlines causes the robot to turn out from a planned path and damage some crops, Navigation controller for an autonomous weed-killer robot

Types of real time tasks

- *A **Periodic task** is executed repeatedly at **regular time intervals**, and each invocation is called a **job or instance**.*
  - *Often time-driven*
- ***A Aperiodic task** is executed to response to **external events**, and to respond, it executes aperiodic jobs whose release time are not know at priori.*
  - *Often event-driven*
  - *Off-line guarantee of aperiodic tasks must make proper assumptions on the environments; that is, by assuming a maximum arrival rate for each event (i.e., minimum inter arrival time)*
- ***Aperiodic tasks** characterized by a **minimum inter arrival time** are called **sporadic tasks**.*

Modeling timing constraints

- *It is a vital attribute in real time system.*
- *It decided the total correctness of the result in real times system.*



Model of real time system

- Real time system is a system that is used for performing some specific tasks.
- These tasks are related with **time constraints** and need to be completed in the time interval.
- The basic model of a real time system presents the overview of all the components involved in a real time system.

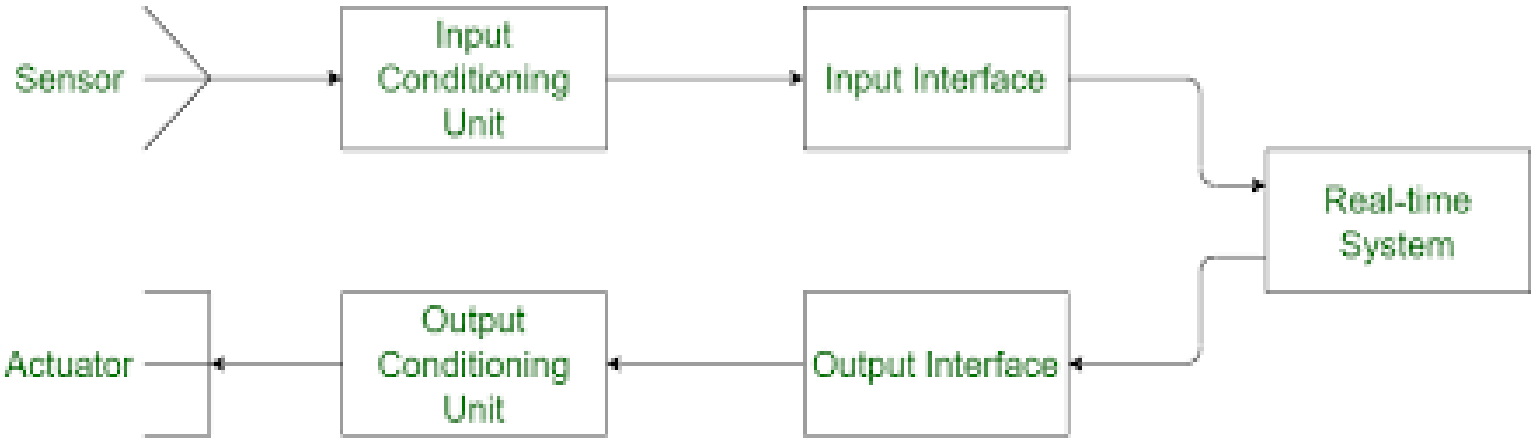
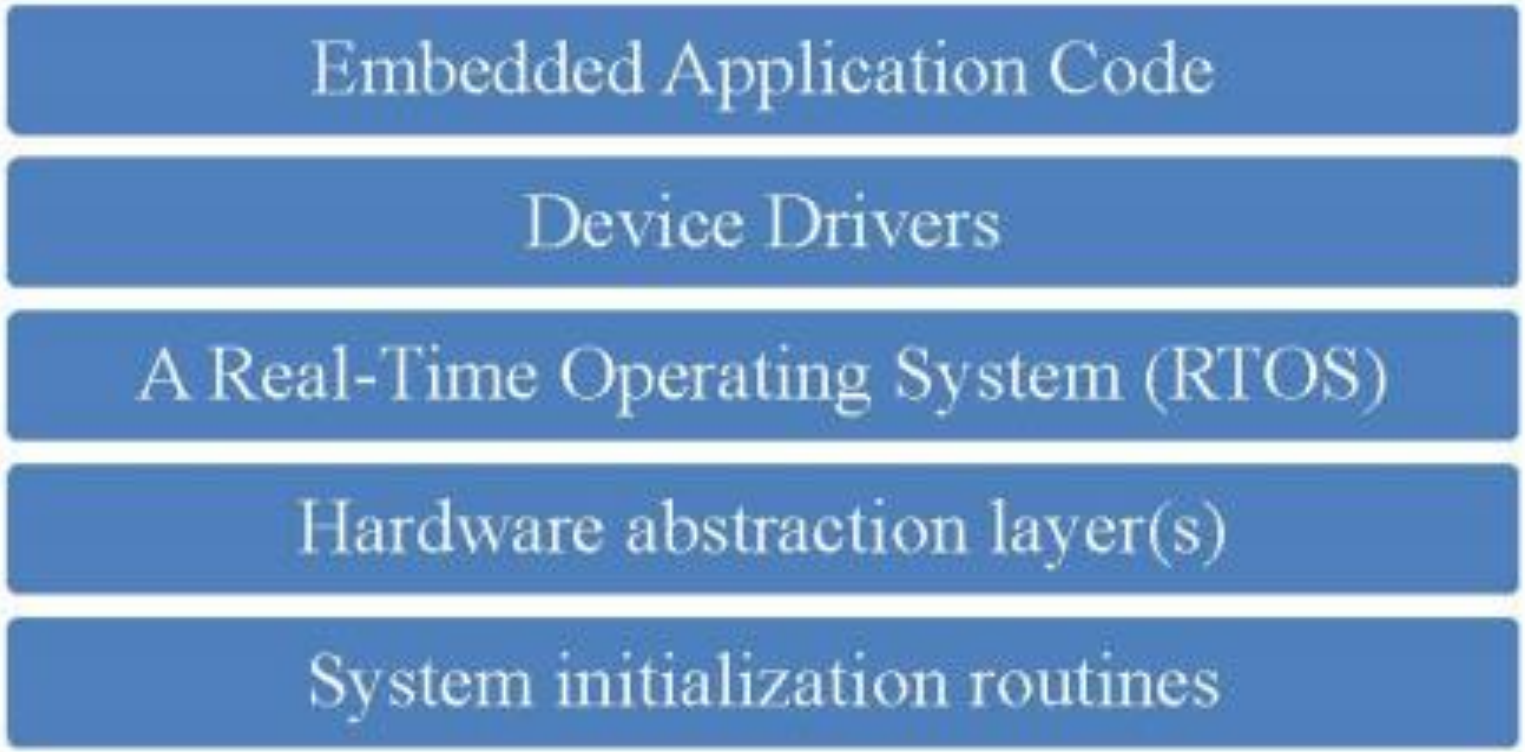
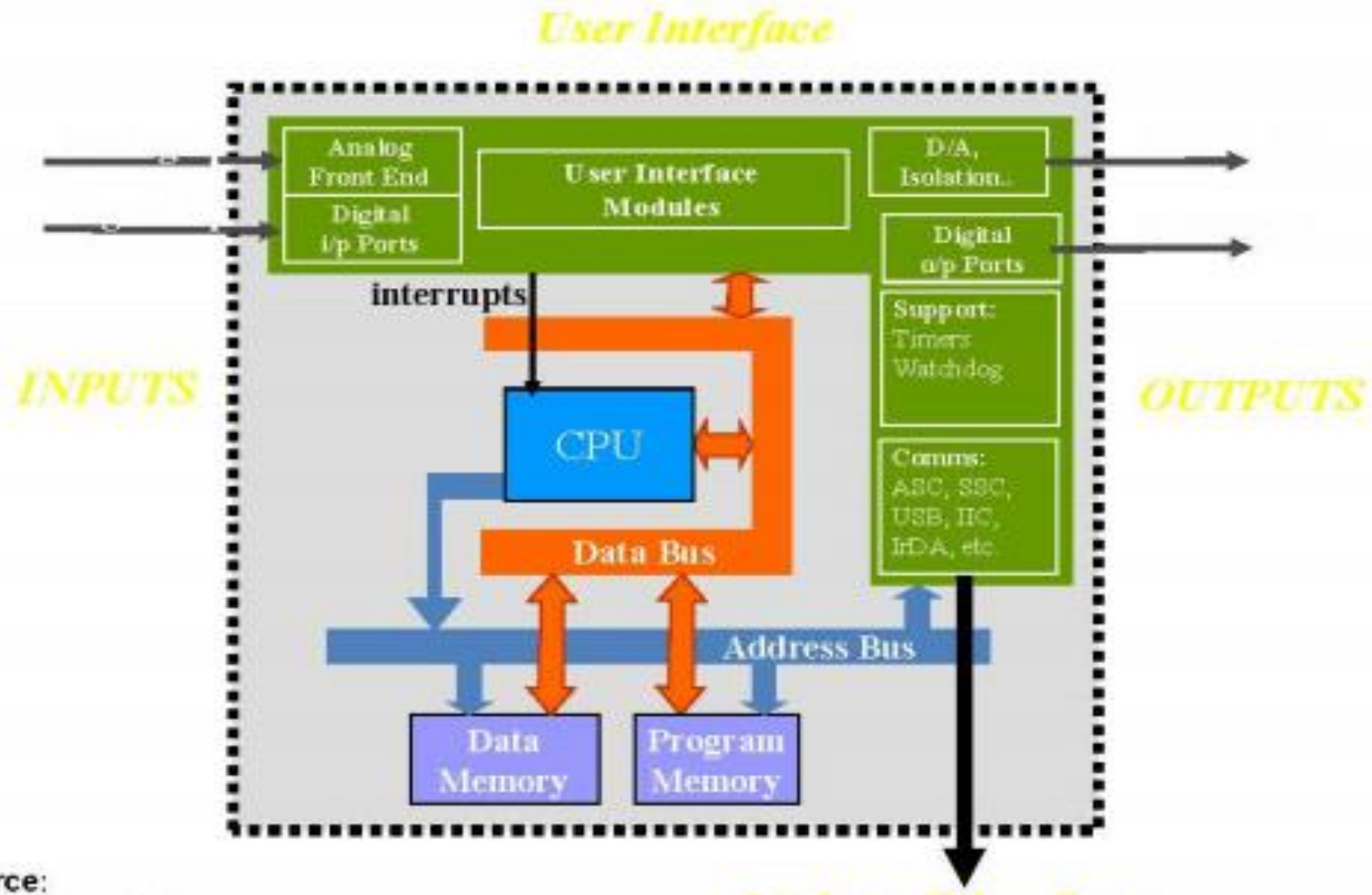


Figure of model of real time system

Typical Embedded System Components

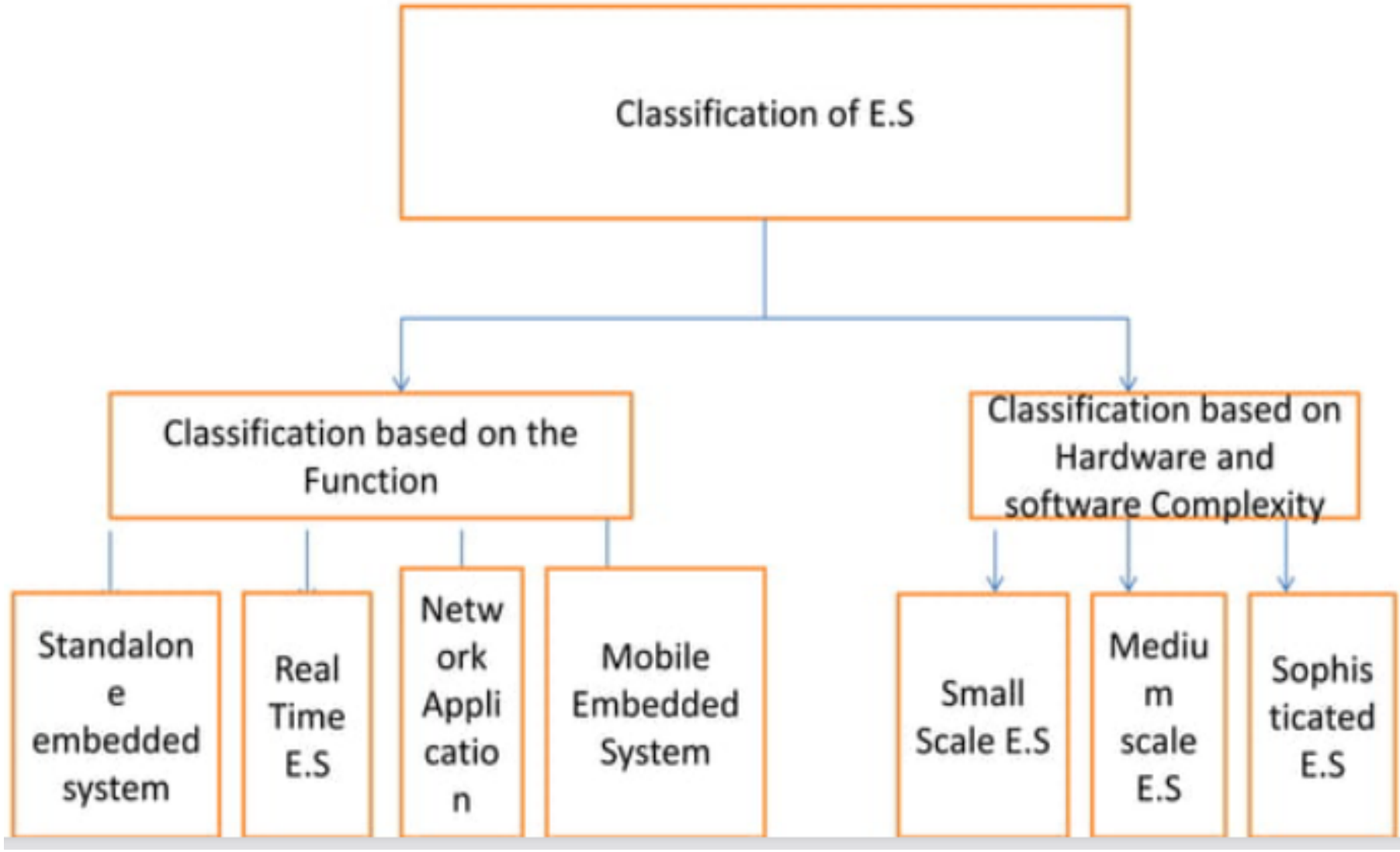


Cntd…



Source:
Sisira K. Amarasinghe,
Verification of High Performance Embedded Systems

Categorization of Embedded Systems



Cntd…

- Embedded systems can be categorization based on **the complexity** in building, **cost factors**, purpose of the system ,**tools** and other related

	Small Scale Embedded System	Medium Scale Embedded System	Sophisticated Embedded System
Processor	Here it can be 8 bit or 16 bit processor. (it can't do computationally intensive process)	It can be a 16 bit or 32 bit processor here. (think of little complex and intensive process with this processor)	PLA, PAL, FPGA and ASIC will fall in this category. (These are high end design elements that can be used to do wonders)
Hardware Complexity	Very little complexity will be visualized here.	We will have to face more complexity in terms of peripheral additions, interfaces etc.,	Highly Complex. (designers need enormous expertise to do proceed with this)

Cntd…

Software Complexity	No complexity will be there in this coding. Because the device is not meant for performing complex functionalities.	There will be complexity added up. This will have few functions and code might be lengthy.	Yeah. Most Complex. Designer needs to be a master to work on the code.
Power	Battery operated	Battery operated	Can be Battery or Live Source based on the application.
Tools Availabiltiy	Can be programmed in simple environment. So no much research on tools is required here.	Here we will have to use Debugger, Compiler, IDE etc., as the task goes slightly cumbersome.	Designer needs sophisticated environment here.
Examples	Calculator can be the simplest example. Stepper motor controller can be added to the list.	Washing Machine, Microwave Oven, and Vending machine.	Flight Landing Gear Systems, Car Braking Systems, Military Applications, Robots.

Cntd…

- **Based on functionality and performance requirements**, embedded systems are also classified as:

1. **Stand alone embedded system:** the software is embedded in the hardware but not necessary the platform of RTOS. It works in a stand alone.
2. **Network Appliances: s**pecific purpose multiple(more than one) hardware those are hosting and meant for specific purpose might need to communicate with one another through network, hence such appliance might need to communicate to one another over a certain network. E.g. IoT.
3. **Real Time embedded system:** Where the software is embedded in the hardware system running over the RTOS platform
4. **Mobile Devices: o**ur mobiles are turning to be one

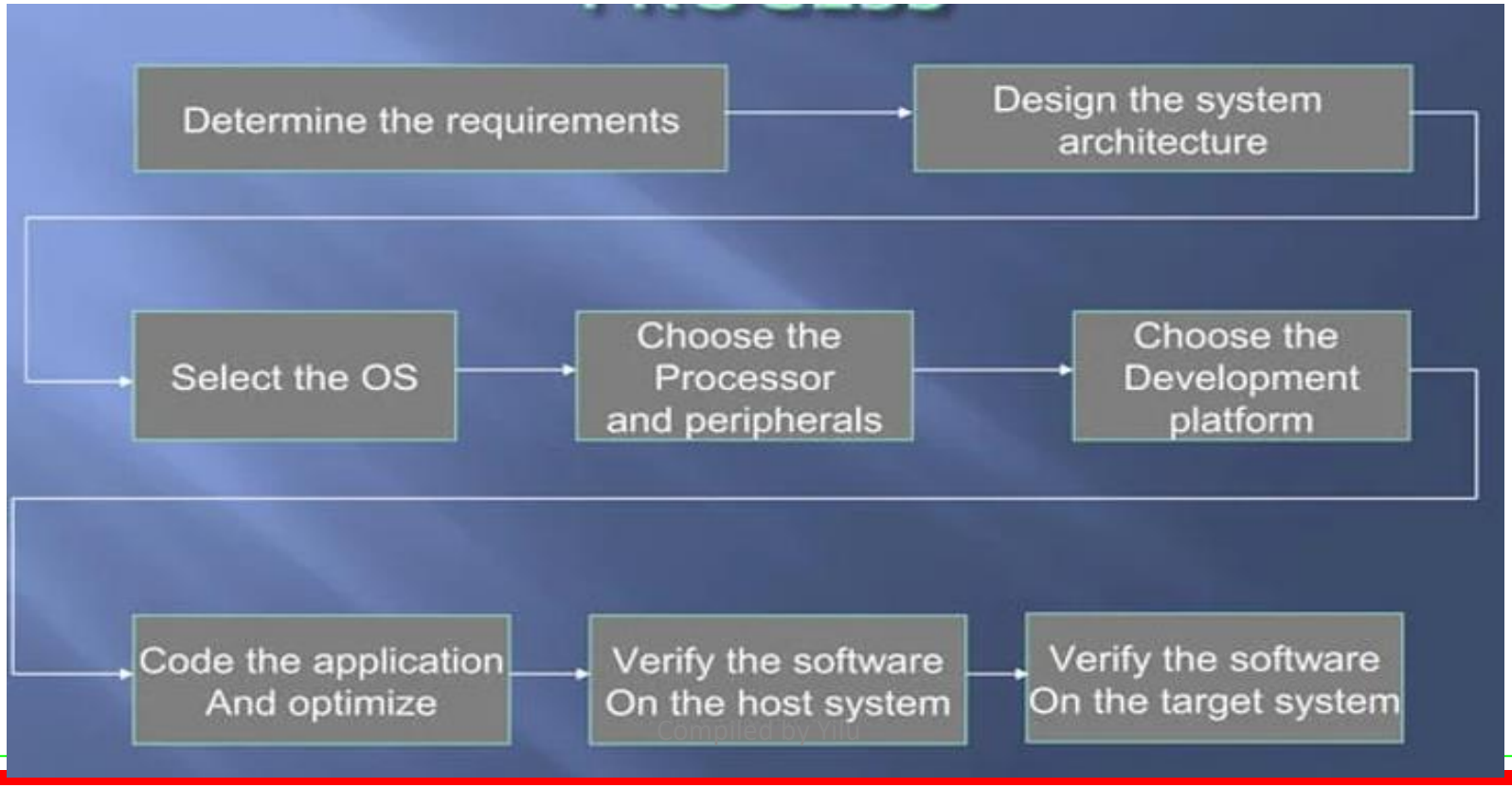
Purpose of Embedded System

- Each embedded system is designed to serve the purpose of any one or combination of the following tasks:

1. Data collection / Storage /Representation E.g. Digital Camera
2. Data Communication E.g. Network hubs, Routers, Switch
3. Data (signal) processing E.g., Digital Hearing aid
4. Monitoring E.g. Electro Cardiogram machine, Multi meter
5. Control E.g. Air conditioner system
6. Application specific user interface E.g. Mobile Phone

Design process

- An embedded system design process is how a manufacturer determines the requirements for a small computerized system embedded within a product.
- Then, they decide the best way to build that system and test that it works.
- The different steps in the embedded system design flow/flow diagram include the following



Development Process

- The development process of embedded system has used four frame works to group the various types of work considered necessary in embedded software development:
  - System engineering process
  - Software engineering process
  - Safety engineering process
  - Support process

ES development process comparison with SDLC /SPM(Assignment)

End of Chapter One

Any Question !!!